M534H HOMEWORK– Spring 2026

Prof. Andrea R. Nahmod

• Set 1. Due date: 02/12/2026

Section 1.1: 2, 3, 4.

Note Note that an equivalent manner (that what we discussed in Lecture 1) of viewing linearity is explained in Strauss' book page 2.

Assigned Background Reading: Review carefully Appendices A1. A2 and A3 in Strauss' book. We will need and use this material through the course. You will also need it to understand the first three Handouts/Section 1.3 of Strauss.

Problem: Do but do not turn in: Prove the *Second Vanishing Theorem* in A.1 page 416

• Set 2. Due date: 02/17/2026

Section 1.1: 11, 12.
Section 1.2: 1, 2, 3, 4, 5, 6.

Additional Problem 1: Solve the transport equation $5u_x - 6u_y = 0$ together with the auxiliary condition that $u(x,0) = 4x^3$.

Additional Problem 2: Solve the inhomogeneous transport equation $2u_x + 3u_y = 1$.

Assigned Reading for next week: Read carefully Handouts 1, 2 and 3 posted in the webpage. These correspond to Examples 2, 3 and 4 in Strauss' Section 1.3